

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

17 September 2018

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Section A:

Answer **ALL** questions in the spaces provided.

Section B:

Answer any **ONE** question.
Write your answers on the separate writing papers provided.

Please hand in section A and section B separately.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 30
2	/ 10
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

Section A

Answer **all** questions in this section.

- 1 Fig. 1.1 shows a molecule of haemoglobin. The haem group plays an important role in the function of haemoglobin.

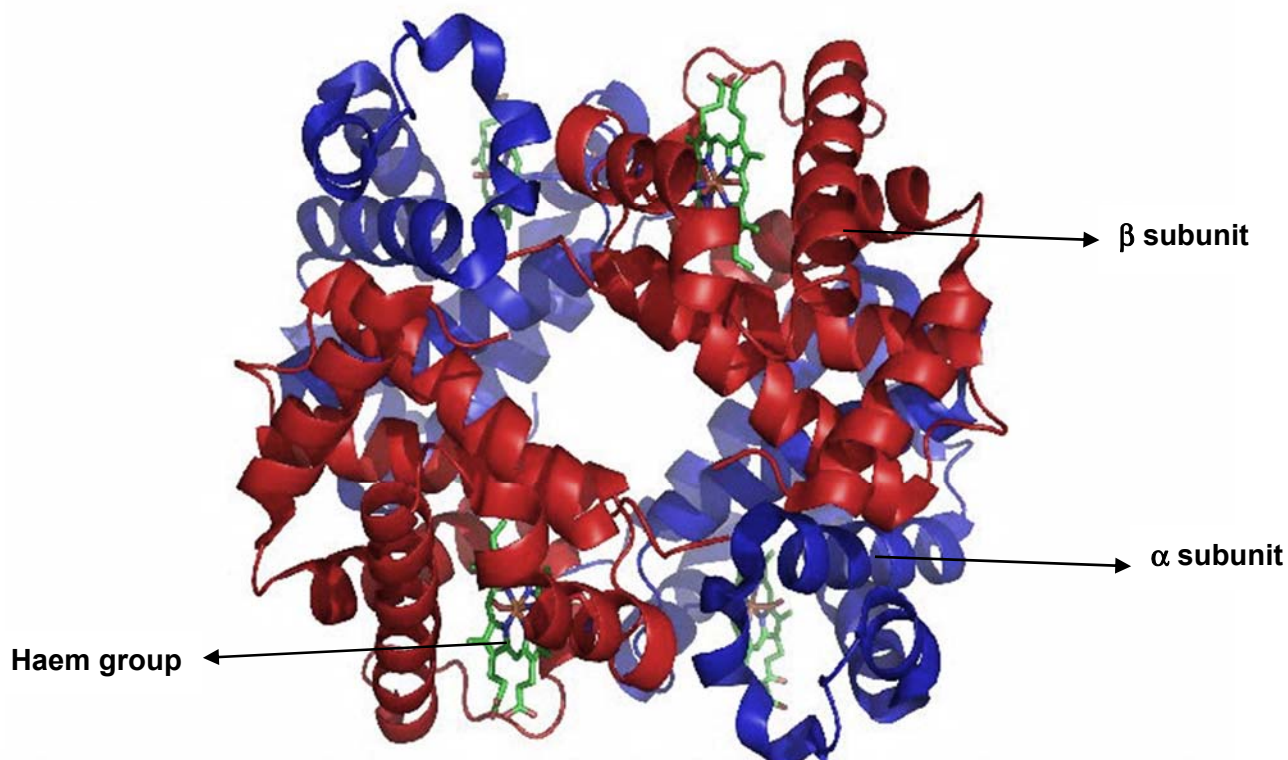


Fig. 1.1

- (a) With reference to Fig. 1.1 and your knowledge, describe **two** structural differences between tropocollagen and haemoglobin. [2]

Tropocollagen	Haemoglobin
Tropocollagen is made up of 3 loose helices / helical strands	Haemoglobin is made up of 2 α -globin and 2 β -globin chains/ 4 sub-units / 4 polypeptide chains
Fibrous, long, strand-like structure	Compact, highly folded, globular / spherical structure
Tropocollagen has hydrophilic R groups on inside and hydrophobic R groups on outside of molecule;	Haemoglobin has hydrophobic R groups on inside and hydrophilic R groups on outside of molecule;
Interchain hydrogen bonds between the polypeptide chains	Hydrogen bonds, ionic bonds, hydrophobic interactions and disulfide bonds-between the polypeptide chains

1 mark for each correct comparison (max. 2)

- (b) Discuss the advantages of having four subunits in haemoglobin. [3]
- a) **Binding of 4 oxygen molecules per haemoglobin results in increased oxygen carrying capacity / more efficient transport of oxygen molecules to respiring tissues;;**
 - b) **Binding of oxygen molecule (to 1 haem group) in 1 subunit changes the 3D conformation of that subunit, which causes change in 3D conformation of the other subunits;;**
 - c) **making it easier for the other 3 subunits to pick up oxygen - ref. to cooperative binding;;**
- (c) Myoglobin is a protein found in the muscle cells of animals. It functions as an oxygen-storage unit, providing oxygen to the working muscles.

Fig. 1.2 shows a molecule of myoglobin.



Fig 1.2

Fig. 1.3 shows the oxygen dissociation curves for myoglobin, fetal haemoglobin and adult haemoglobin.

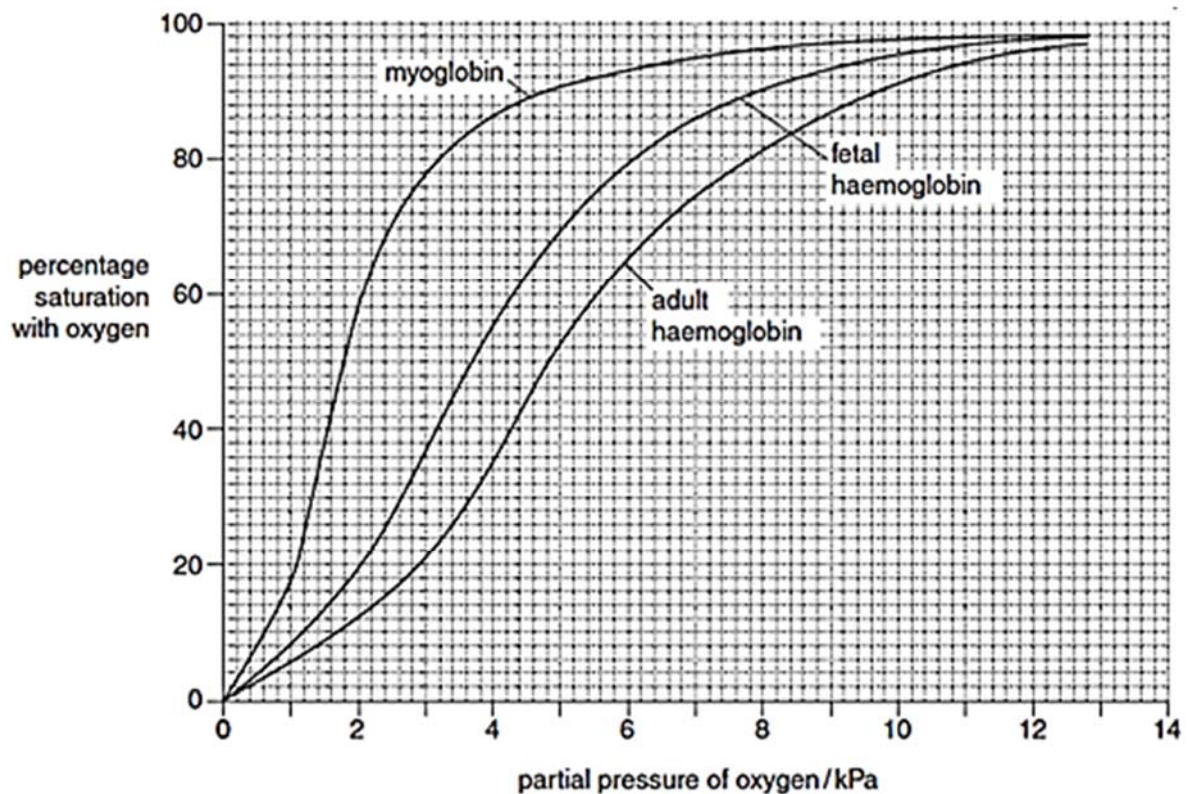


Fig. 1.3

- (i) Name the cells in which haemoglobin is found. [1]
- red blood cells / erythrocytes;;**
- (ii) Use Fig. 1.3 to determine the percentage saturation of myoglobin and adult haemoglobin when the partial pressure of oxygen is 3 kPa. [1]
- myoglobin 77 or 78%, adult haemoglobin 21%;;**
- (iii) There is a large difference between the percentage saturation of myoglobin and that of adult haemoglobin at low partial pressures of oxygen. Suggest reasons for this. [2]
- a) **Myoglobin has higher affinity for oxygen/myoglobin binds oxygen while haemoglobin releases oxygen;;**
 - b) **Myoglobin acts a store of oxygen;;**
 - c) **Myoglobin will only release oxygen at (very) low oxygen partial pressure/ AW when oxygen demand (in muscles) exceeds supply;**
 - d) **AVP - e.g. myoglobin has one / fewer haem groups, so no cooperative binding effects, e.g. allows aerobic respiration to continue (in muscles);;**
- max. 2**

- (d) Fetal haemoglobin has a different oxygen binding affinity to that of adult haemoglobin, as shown in Fig. 1.3. Normally, after birth, the production of fetal haemoglobin stops and the adult form is produced.

In a rare condition known as Hereditary Persistence of Fetal Haemoglobin (HPFH), fetal haemoglobin continues to be produced well into adulthood in addition to adult haemoglobin. This condition, however, usually lacks any symptoms.

- (i) Explain, with reference to Fig. 1.3, the significance of the difference in oxygen binding affinity between fetal and adult haemoglobin. [2]

a) At all ppO_2 , fetal haemoglobin has higher oxygen affinity (than adult haemoglobin) /AW;;

b) quote data at more than one ppO_2 (from Fig. 1.1);;
max. 1

c) so that oxygen uptake from maternal blood/AW can take place;;

d) so that gas exchange between fetal and maternal blood can take place;;

e) ref. to fetal reliance on mother to supply oxygen / mother only source of oxygen for fetus;

max. 1

- (ii) Suggest why HPFH usually lacks symptoms. [1]

a) at lower ppO_2 both unload/AW oxygen;;

b) sufficient/more adult haemoglobin present / adult haemoglobin provides sufficient oxygen;;

c) ref. to compensating by producing additional red blood cells;;

d) AVP - e.g. ref. to similarity of position of both oxygen dissociation curves;;

max. 1

Answer must have idea of comparing adult haemoglobin with fetal haemoglobin

Haemoglobin disorders are inherited blood diseases that affect the quality or amount of haemoglobin and the capacity to carry oxygen around the body. They fall into two main categories: thalassemia and sickle cell disease.

β -thalassemia is caused by mutations in the β -globin gene. People with severe thalassemia depend on blood transfusions to give them working red blood cells. A type of bone marrow transplant has been used to try to cure thalassemia, but it is hard to find compatible donors. Even if a donor can be found, there is still a risk that the patient's body will reject the transplant.

- (e) Explain why a bone marrow transplant from a healthy donor can be used to cure thalassemia. [2]

a) HSCs in the bone marrow are multipotent, unspecialised cells, that divide and differentiate into closely related family of cells including RBCs and WBCs / lymphocytes;;

b) HSCs from donor do not have the mutation causing thalassemia, thus upon differentiation, they give rise to normal / working RBCs which will replace worn out recipient RBCs;;

- (f) β -thalassemia major is a severe form of β -thalassemia requiring lifelong blood transfusion. Cardiac complications are the major cause of death in patients with β -thalassemia major. Studies have shown that a non-competitive enzyme inhibitor improves cardiac functions in blood transfusion-dependent patients with β -thalassemia major.

(i) Explain the effects of a non-competitive inhibitor on the rate of enzyme activity. [3]

- a) reduces rate of enzyme activity;;
 - b) by binding to a site on the enzyme other than at the active site / allosteric site, causing change in 3D conformation / configuration of active site;;
 - c) substrate unable to bind to enzyme active site / ES complexes do not form / fewer ES complexes / products do not form;;
 - d) AVP, e.g. V_{max} not reached / increasing substrate concentration no effect;;
- max. 3

Fig. 1.4 shows two ways enzymes interact with their substrates.

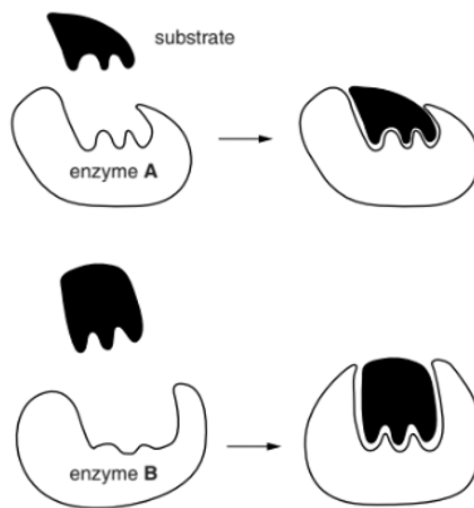


Fig. 1.4

(ii) Explain the difference between the **two** ways in which enzymes interact with their substrates as shown in Fig. 1.4. [2]

- a) Enzyme A uses 'lock and key' and enzyme B uses induced fit;;
 - b) Enzyme A / lock and key, (shape of) active site is complementary/AW to (shape of) substrate (molecule);;
 - c) Enzyme B / induced fit, has an active site that moulds around/AW the substrate;;
- max. 2

Sickle cell anemia is an autosomal, recessive human disease. A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in Table 1.1.

Table 1.1

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

(g) Based on the information in Table 1.1,

(i) which patient(s) has / have the sickle cell anemia mutation? [1]

Patients 3 & 5;;

(ii) explain how the change of amino acid would result in a sickle shaped red blood cell. [3]

- a) tertiary structure of β -globin chains in haemoglobin changed, as valine that has a hydrophobic R group has replaced glutamic acid that has a hydrophilic R group;;**
- b) effect of mutated haemoglobin occurs at low oxygen concentration, where hydrophobic regions on haemoglobin molecules stick together / solubility of deoxygenated haemoglobin decreases;;**
- c) haemoglobin molecules polymerise/crystallise into long fibres, causing RBCs to adopt a sickle shape;;**

Fig. 1.5 shows the distribution of malaria and the allele frequency of the sickle cell allele in Africa.

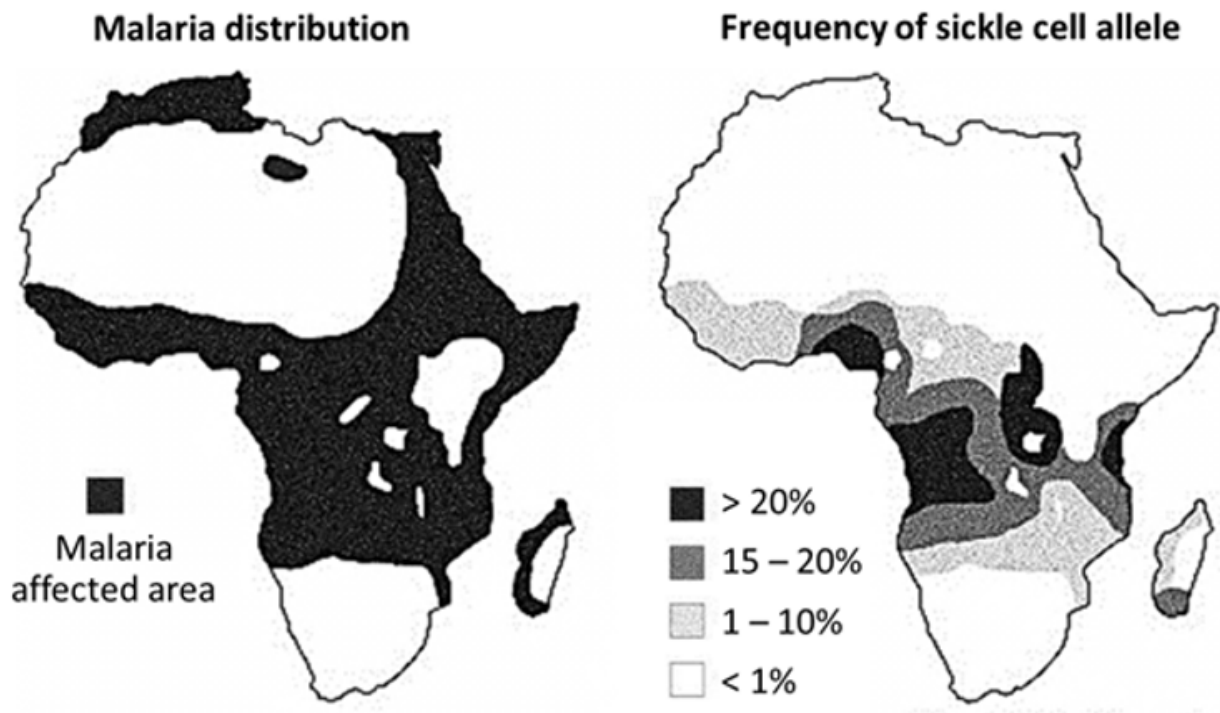


Fig. 1.5

(h) Explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele. [3]

- Frequency of sickle cell allele is higher in malaria affected areas;;
- As heterozygous individuals do not exhibit sickle cell anaemia / are phenotypically normal, they are at a selective advantage / there is heterozygote advantage in malaria affected areas;;
- Heterozygous individuals are more likely to survive to reproduce and pass on the sickle cell allele to viable and fertile offspring;;
- hence increasing the allele frequency of the sickle cell allele in the population and change in gene pool within these malaria areas over time;;

max. 3

The genotypes of five different individuals living in Central Africa are identified by Restriction Fragment Length Polymorphism (RFLP) analysis, using the restriction enzyme *MstII*. Individuals may have the normal allele for the β -globin gene, the sickle cell allele or both alleles.

The β -globin gene sequence in humans is flanked by two *MstII* restriction sites, and in the normal allele, there is another *MstII* restriction site within the gene sequence.

The autoradiograph obtained from the RFLP analysis is shown in Fig. 1.6.

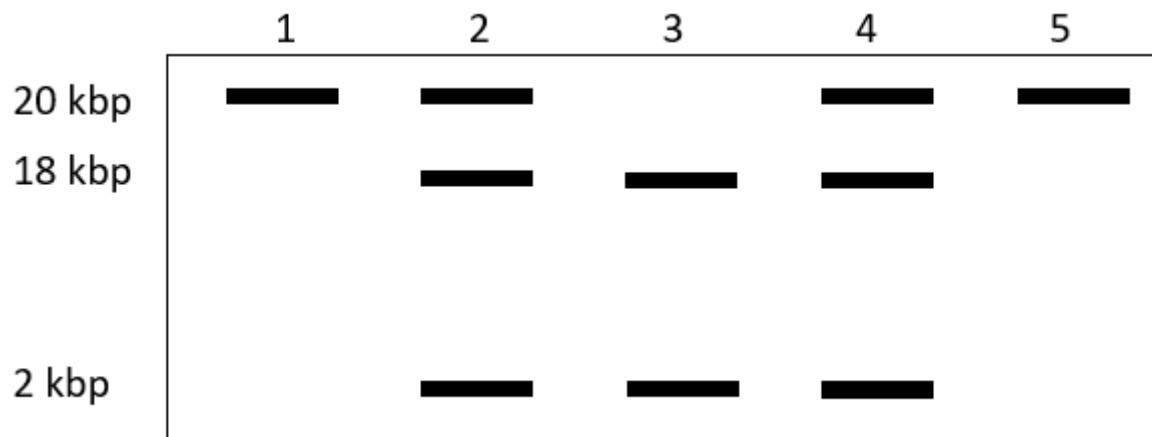


Fig. 1.6

- (i) With reference to Fig. 1.6, identify and explain which individuals are carriers of the sickle cell disease. [4]
- a) Individuals 2 & 4 are carriers;;
 - b) When the normal allele is cut with *MstII*, two DNA fragments: 18 kbp and 2 kbp, are obtained;;
 - c) For the sickle cell allele, mutation in the DNA sequence has resulted in the loss of the *MstII* restriction site within the DNA sequence (of the normal allele), so only 1 DNA fragment, 20 kbp, is obtained when cut with *MstII*;;
 - d) Individuals who are carriers for sickle cell anemia have 1 copy of the normal allele and 1 copy of HbS allele, so autoradiograph will show 3 bands: 20 kbp, 18 kbp and 2 kbp;;

[Total: 30]

- 2 The *lac* operon is involved in the transport and metabolism of lactose in *Escherichia coli*. The operon is regulated by molecules that turn gene expression on and off in response to the concentration of nutrients.

- (a) Describe how the concentration of glucose in a bacteria cell regulates the expression of the *lac* operon in the presence of lactose. [4]

In the presence of lactose,

Low levels of glucose (max 2)

- a) high concentration of cyclic AMP/cAMP (in the cell) and cAMP binds to allosteric site of catabolite activator protein;;
- b) 3D conformation of CAP changes to active conformation, active CAP binds to CAP binding site within the lac promoter;;
- c) which increases affinity of RNA polymerase to the promoter, increasing transcription rate of structural genes;;

High levels of glucose (max 2)

- a) low concentration of cyclic AMP / cAMP;;
- b) CAP remains in its inactive conformation and inactive CAP unable to bind CAP binding site;;
- c) RNA polymerase binds weakly to the promoter, transcription rate is low / at basal level since lac repressor does not bind to operator in the presence of lactose;;

- (b) Suggest and explain the importance of operons in bacteria. [2]

- a) allow bacteria to respond to environment appropriately under different conditions;;
- b) thus bacteria able to adapt to changes in the environment;;

OR

- c) conserve resources by producing the enzymes / proteins only when required;;
- d) gives bacteria cells a selective advantage since they are able to prevent the waste of energy and resources;;

- (c) Explain why the mRNA encoded by the *lac* operon is described as polycistronic. [2]

- a) mRNA of the lac operon contains the mRNA sequences of three different polypeptides separated by three start codons and three stop codons;;
- b) upon translation, three different polypeptides are produced from one single mRNA;;

Fig. 2.1 shows a hybrid operon where the structural genes of the *trp* operon have been fused by deletion to the *lac* operon. The genes necessary for the biosynthesis of the essential amino acid tryptophan remain intact.

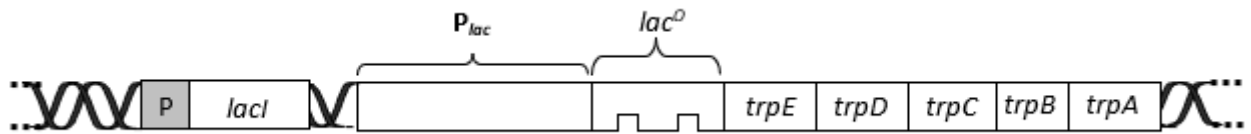


Fig. 2.1

(d) Predict the amount of EDCBA proteins by filling the table below using symbols:

“-” → negligible protein product

“+” → protein product

“++” → lots of protein product

Metabolites present

Glucose	-	+	-	-	+	+	-	+
Lactose	-	-	+	-	-	+	+	+
Tryptophan	-	-	-	+	+	-	+	+
Amount of EDCBA proteins	-	-	++	-	-	+	++	+

At least 4 boxes → 1m

All 8 boxes correct → 2m

[2]

[Total: 10]

- 3 The dengue virus is a member of the genus *Flavivirus* in the family *Flaviviridae*. Along with the dengue virus, this genus also includes a number of other viruses transmitted by mosquitoes and ticks that are responsible for human diseases. *Flavivirus* includes the yellow fever, West Nile, Japanese encephalitis, and tick-borne encephalitis viruses.

The dengue virus is particularly notable in that it exists as four antigenically distinct serotypes (denoted as DENV-1 to DENV-4), within which there is considerable genetic variation in the guise of phylogenetically defined “genotypes”.

- (a) Describe how the dengue virus develops in humans. [3]

Cellular level:

- a) virus enters body through a bite by an infected mosquito and binds to receptors found on keratinocytes and tissue-resident Langerhans cells and macrophages;;
- b) uptake of virus through receptor-mediated endocytosis and acidification of endocytic vesicle causes viral envelope to fuse with vesicle membrane;;
- c) release of viral RNA into cytoplasm, where host cell machinery is used to replicate the virus and release of virus out of cell;;

Physiological level:

- d) infected Langerhans cells gets activated, releasing cytokines to recruit circulating monocytes to site of infection, resulting in increased availability of host cells, hence increasing viral replication;;
- e) further increase in viral replication also occurs when infected Langerhans cells / macrophages migrate through the lymphatic vessels, leading to viremia;

max. 3

- (b) Describe **two** ways in which the production of antibodies helps in removing DENV from the body in the primary antibody response. [2]

- a) Neutralisation, where the binding of the antibodies to the antigen prevents the interaction of the DENV to the host cell receptors;;
- b) Opsonisation, where the antibody-bound DENV is recognised by phagocytes, thus enhancing the clearance of the DENV from the blood;;
- c) Agglutination, where DENV bound by antibodies are concentrated, hence lesser infectious units and easier to clear DENV;;
- d) Activation of complement proteins, where the antibody-bound DENV recruits complement proteins that assemble on DENV surface, promoting lysis;;

max. 2

- (c) Explain why an individual may not be necessarily immunised against a second dengue infection. [2]

Global warming is the unusually rapid increase in Earth’s average surface temperature over the past century primarily due to the greenhouse gases released as a result of anthropogenic activities. The global average surface temperature rose by 0.6 to 0.9°C between 1906 and 2005, and the rate of temperature increase has nearly doubled in the last 50 years. Temperatures are certain to go up further.

(d) Explain how global warming has encouraged the spread of dengue. [3]

[Effect of temperature]

- a) Higher temperatures hasten the life cycle of mosquitoes due to increased metabolism, hence producing more offspring;;
- b) Higher temperatures cause female mosquitoes to feed more frequently due to increased rate of digestion, increasing transmission intensity;;
- c) Temperate countries are now experiencing a warmer temperature, thus encouraging mosquitoes to migrate to higher latitudes;;

[Effect of precipitation]

- d) Global warming causes increased precipitation, hence increasing the number of breeding sites for mosquitoes;;

max. 3

[Total: 10]

Section B

Answer **ONE** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clear labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a)** Describe the production and folding of a functional enzymatic protein that is used within a cell. [13]

Describe transcription: (max 4)

- a) general transcription factors and RNA polymerase (II) bind to promoter, forming the transcription initiation complex;;
- b) RNA Polymerase II unzips and unwinds the DNA double helix;
- c) RNA polymerase (II) aligns ribonucleoside triphosphates / free ribonucleotides with the DNA template strand according to complementary base-pairing rules: A-U, T-A, G-C, C-G;;
- d) RNA Polymerase (II) joins the ribonucleotides together by catalysing the formation of phosphodiester bonds;;
- e) growing mRNA is synthesised in the 5' to 3' direction;;
- f) when RNA polymerase (II) reads the termination signal (in prokaryotes), transcription ends
OR
when RNA polymerase (II) transcribes past polyadenylation signal (in eukaryotes), transcription ends;;

Describe post-transcriptional modification (in eukaryotes): (max 2)

- g) pre-mRNA (in eukaryotes) undergoes post-transcriptional modification;;
- h) addition of a 7-methylguanosine cap (5' methylated guanosine triphosphate cap) at 5' end of mRNA, catalysed by capping enzyme complex;;
- i) under the action of spliceosomes, splicing occurs where introns are removed/excised and exons are spliced together;;
- j) addition of a poly-A tail to the 3' end of the mRNA, catalyzed by poly A polymerase;;
- k) mature mRNA leaves nucleus through nuclear pores and enter cytoplasm;;

Describe translation: (max 5)

- l) initiation of translation at 5' end of mRNA involves assembly of small and large ribosomal subunits and initiator tRNA;;
- m) initiator tRNA is at the P site, anticodon of initiator tRNA complementary base pairs with start codon AUG;;
- n) second aminoacyl-tRNA complex with anticodon complementary to next codon is brought into the A site;;
- o) amino-acyl tRNA complex is a tRNA molecule that carries a specific amino acid at its 3' end, depending on the specific anti-codon, formation of this complex is catalyzed by specific aminoacyl-tRNA synthetases;;
- p) peptide bond forms between the amino acid on the tRNA in the A site with the amino acid/polypeptide bound to tRNA in P site, this is catalysed by peptidyl transferase;;
- q) ribosome moves three bases (a codon) downstream (in the 5' to 3'direction), tRNA in P site is translocated to E site and released;;
- r) elongation continues until ribosome reaches a stop codon (UAA/UGA/UAG);;

- s) release factor binds directly to the stop codon, causes addition of water molecule, hydrolysing the polypeptide from the tRNA and releasing polypeptide from the ribosome;;

Describe post-translational modification: (max 4)

- t) secondary level of structure of protein/enzyme involves hydrogen bonds between CO and NH groups of the peptide bonds / polypeptide backbone, forming structures such as alpha helices and beta pleated sheets;;
- u) enzymes are globular proteins with unique three-dimensional conformation, so polypeptide chain has to fold into this unique conformation, held together by hydrophobic interactions, disulfide, ionic and hydrogen bonds between R groups of amino acids;;
- v) In enzymes, folding must also give rise to an active site that is complementary to the substrate / active site conformation ready to receive the substrate;;
- w) folding brings catalytic amino acids residues far apart in the primary structure / polypeptide close together at the active site;;
- x) cleavage of polypeptide may be needed for the enzyme to be functional;;
- y) certain amino acids may need to be chemically modified by the attachment of chemical groups, e.g. phosphate groups;;

Content: Total - max 12

QWC – transcription, translation and post-translational modification must each be described in a separate paragraph

- (b) The effects of global warming on the spread of malaria beyond the tropics is a debatable issue. Discuss the arguments/**evidences** that support your stance on the matter, and provide a balanced account of counter-arguments. [12]

Arguments/evidences supporting the theory that global warming causes spread of malaria beyond the tropics:

- a) Malaria can only occur in climates where mosquitoes are present to transmit the disease;;
- b) Global warming leads to temperatures in regions beyond the tropics, e.g. sub-tropics, becoming more optimal for both mosquitoes and malaria parasite;;
- c) Higher temperature leads to higher rate of development / metabolic rate / survival of mosquitoes;;
- d) As mosquitos are able to mature faster;;
- e) Global warming can lead to higher humidity (in the sub-tropics), more stagnant pools of water available for mosquitoes to lay eggs;;
- f) Effects of temperature and climate on malaria have led many to believe that global warming will result in the spread of this disease, to higher latitudes (sub-tropical regions) where it used to be cooler;;
- g) AVP;;

Arguments and evidences against the theory that global warming causes spread of malaria beyond the tropics:

- h) Common assumption that a faster development of plasmodium parasite leads to higher rate of transmission, but this may not be so, As temperature rises, parasites do develop faster, but fewer of them become infectious;;
- i) Even if an increase of temperature leads to higher parasite development, this may lead to a decrease in malaria risk;;
- j) This is because the parasite may not be able to cope with the higher temperatures or
- k) because mosquito immune system may work better at warmer temperatures;;
- l) Need to take into consideration other more significant factors in the recent upsurge of malaria;;
- e.g.
- m) Poor vector control - less use of DDT / insecticides, resulting in higher proliferation of mosquitoes;;
- n) Changes in land use - extension of villages of towns, deforestation, building of dams, human populations coming closer to the wilderness and hence in closer proximity to mosquito breeding grounds;;
- o) Population growth – may result in increase in malaria, if there is no simultaneous improvement in healthcare facilities and living conditions;;
- p) Migration and human travel can cause the spread of malaria from one area to another (where the people have low immunity to the disease);;
- q) AVP;;

Content: Total - max 11

QWC: at least 3 points in each side of the argument (for and against)

[Total: 25]

5 (a) All living organisms need to synthesise ATP. Summarise how cells produce ATP. [13]

- a) ATP molecule is made up of ribose sugar, adenine, and three phosphate groups;;
- b) ATP is formed from ADP by the addition of an inorganic phosphate (P_i) by condensation, this process is also known as phosphorylation;;

In aerobic respiration,

- c) during glycolysis, in the process of converting glucose to pyruvate, 2 net ATP molecules per glucose molecule are produced via substrate level phosphorylation;;
- d) during the Krebs cycle, when succinyl coA is converted to succinate, 2 ATP molecules per glucose molecule is produced via substrate level phosphorylation;;
- e) glycolysis, link reaction yields NADH, and Krebs cycle yields NADH and $FADH_2$, which act as carriers of protons and electrons to the electron transport chain (ETC);;
- f) where most of the ATP is synthesised during oxidative phosphorylation;;

Oxidative phosphorylation in respiration: (max 5)

- g) During oxidative phosphorylation in respiration, synthesis of ATP occurs at the inner mitochondrial membranes by chemiosmosis;;
- h) NADH and $FADH_2$ transfer protons and high energy electrons to the ETC;;
- i) Each NADH contains sufficient energy to produce 2.5 ATP molecules following oxidation, each $FADH_2$ contains sufficient energy to produce 1.5 ATP molecules following oxidation;;

Mark once, either in oxidative phosphorylation or photophosphorylation:

- j) in ETC, electrons are transported along a series of electron carriers which are of progressively lower energy levels;;
- k) energy released from electron transport is used to pump protons from the mitochondrial matrix, across the inner mitochondrial membrane, into the intermembranal space;;
- l) creating a proton gradient across the inner mitochondrial membrane, which is a source of potential energy for the synthesis of ATP, also known as a proton motive force;;
- m) ATP is synthesised from ADP and P_i when protons move down proton gradient from the intermembranal space into the mitochondrial matrix (by facilitated diffusion) through the channels in ATP synthase;;

During anaerobic respiration,

- n) glycolysis and fermentation take place;;
- o) fermentation serves to regenerate NAD^+ from NADH for glycolysis to continue to take place to produce ATP until oxygen becomes available again;;
- p) As link reaction, Krebs cycle and oxidative phosphorylation do not take place in the absence of oxygen, only glycolysis occurs, producing only 2 molecules of ATP per molecule of glucose via substrate level phosphorylation;;

Photophosphorylation in photosynthesis (max 5)

- q) During the light dependent reaction of photosynthesis, synthesis of ATP occurs at the thylakoid membranes by chemiosmosis;;
- r) In non-cyclic photophosphorylation and cyclic photophosphorylation, light energy causes photoactivation of photosystems II & I, electrons from the reaction centres/photosystems are being passed to ETC;;

Mark once, either in oxidative phosphorylation or photophosphorylation:

- s) in ETC, electrons are transported along a series of electron carriers (plastoquinone, cytochrome complex and plastocyanin) which are of progressively lower energy levels;;
- t) energy released from electron transport is used to pump protons from the stroma, across the thylakoid membrane, into the thylakoid space;;
- u) creating a proton gradient across the thylakoid membrane, which is a source of potential energy for the synthesis of ATP, also known as a proton motive force;;
- v) ATP is synthesised from ADP and P_i when protons move down proton gradient from the thylakoid space into the stroma (by facilitated diffusion) through the channels in ATP synthase;;

Content: Total - max 12

QWC – describe at least 2 out of 3 processes on how cells produce ATP, with clear paragraphing between each process

(b) Discuss why life would be impossible without ATP.

[12]

- a) Characteristics of life include growth, reproduction, excretion, movement (in animals), taking in substances from the environment (food in animals and CO₂ and mineral salts in plants), all these require ATP;;
- b) ATP makes an important contribution to an efficient energy conversion through respiration at room temperature, without which organisms would not have sufficient energy to maintain life without over-heating and denaturing proteins / enzymes;;
- c) Without ATP, light independent reactions of photosynthesis cannot take place, plants would not grow and animals would have no food since all organic matter is derived ultimately through photosynthesis (through food chains);;
- d) ATP is needed in the Calvin cycle, during the reduction phase, where each molecule of 3-phosphoglycerate receives an additional phosphate group from ATP, forming 1,3-bisphosphoglycerate as a product;;
- e) in the Calvin cycle, ATP is also need for the regeneration of RuBP from G3P;;
- f) Without ATP, animals would be unable to move (other than extremely slowly – when in fact many mammals can run at high speeds) to find/catch food / to escape from predators/harm / to migrate to cope with seasonal environmental changes;;
- g) ATP is needed for muscle contraction, where myosin filaments of muscle fibers slide past the actin (thin) filaments (ref. to sliding filament theory);;
- h) Without ATP, organisms would not be able to grow, reproduce or maintain themselves, ref. to blood cells / enzymes / hormones need to continually replaced;;
- i) deoxyATP, one of the building blocks of DNA, is needed for DNA replication, along with deoxyTTP, deoxyGTP and deoxyCTP;;
- j) in protein synthesis, ATP has a similar role during RNA synthesis / transcription, along with TTP, GTP and UTP;;
- k) cell division could not take place as ATP is required during mitosis, especially during anaphase;;
- l) immune system of humans / mammals need ATP to maintain a turnover of the different types of white blood cells / produce antibodies rapidly and in large quantity / for phagocytes to ingest pathogens;;
- m) Without ATP, plants could not absorb mineral ions from the environment / cells in general could not regulate passage of substances across the membrane;;
- n) ATP is required for carrier proteins to pump substances across membranes via active transport;;
- o) Without ATP, animals could not coordinate their bodies / respond to stimuli / carry out osmoregulation;;
- p) In the nervous system, ion pumps create an action potential and the secretion of neurotransmitters across synapses;;
- q) In the excretory system, kidney function relies on ATP for reabsorption of glucose and salts;;
- r) AVP;; e.g.
 - bioluminescence in glow worms, fireflies, planktonic protocists
 - electrical discharge in some fish
 - beating of cilia and flagella in protocists
 - movement of vesicles round cell on microtubule tracks
 - movement of cytoskeleton / endocytosis and exocytosis

Content: Total - max 11

QWC –describe at least 5 processes on why life would be impossible without ATP, with clear paragraphing between each process

[Total: 25]

END OF PAPER